Improving cell theory

Scientists improve their theories as new evidence is collected. The idea of **cells** changed over a long time as more evidence was found.

- 1665 – Robert Hooke first saw and named cells. Possibly the first theory would have begun as: *There are some cells in living things.*
- 1839 – Matthias Schlieden and Theodor Schwann revised the original idea to read: *All living things are made from cells.*
- Soon after, Schlieden and Schwann decided that the theory should be revised to include a reference to the materials that cells make, such as bone matrix.
- 1858 – Rudolf Virchow added the idea that all cells come from other cells. He is now believed to have copied this idea from Robert Remak.

The theory was then very close to the definition we use today:

All living things are made from cells and their products. All cells come from other cells.

Science is open to change, but the changes will only be accepted if they are very strongly supported and not able to be disproved.

☞ **1** Match these sentence halves to summarise the ideas above.

Living things are made of	lot of cell product around every cell.
Hooke used his improved microscope to observe	but become famous by understanding the importance of what they have seen.
The matrix is the material	cells in cork and to name them 'cells'.
Schlieden and Schwann's theory was improved	cells and cell products. Cells always come from other cells.
Some tissues, such as bone, have a	that surrounds the cell in tough tissues.
Scientists sometimes see things others have seen before	there is evidence for the change and it has not been disproved.
Scientific theories only change when	after many observations and communications between scientists.

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